



SECI1143

PROBABILITY & STATISTICAL DATA ANALYSIS

SECTION 03 GROUP 06

PSDA PROJECT 2

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1.0 Introduction

Our team is utilizing the dataset collected during our first project to proceed with our Project 2. For this current project, we need to perform a statistical inference analysis that includes various analysis based on our dataset. There are a few compulsory analyses including the hypothesis testing (1-sample or 2-sample), correlation, and regression. The title of our recent project is "The Learning Mode and Its Impact on Students' Academic Performance." The objective is to explore students' learning approaches and how their engagement in the curriculum influences their academic results.

In this project, we aim to investigate whether there is a relationship between students study type, study period in a day, CGPA, number of courses taken, satisfaction level for learning mode and comfortability of their study mode. To analyze and interpret this dataset, our group have decided to use the hypothesis 2 sample test, correlation test, regression test, and chi-square test of independence.

We were expecting to see there is slightly difference in mean of study hours between student who study regularly and those who seldom do revision. We are investigating to get the relationship between the study period of students in a day and their CGPA. Furthermore, we were expecting to get the result on students' satisfaction of their learning mode are partially dependent on their study habit. Lastly, we expect the students' study habit are highly dependent on their satisfaction of their own study mode.

2.0 Data Set

Our data set is from the data collection by Project 1 using Google Form. This data set consist of 18 different data collecting questions or variables, but we only use 6 out of 18. The table below show the variables and their level of measurement and type of data:

No.	Question/Variable	Level of Measurement	Type of Data
1.	Study habit	Nominal	Categorical
2.	Study Period in a day (Hour)	Interval	Continuous
3.	CGPA	Interval	Continuous
4.	Number of courses taken	Ratio	Discrete
5.	Comfortability of study mode (1-5)	Ordinal	Categorical
6.	Satisfaction level for learning mode (1-5)	Ordinal	Categorical

Table 2.1: Type of Variables

We choose this dataset because it consists of some factor that will affect the academic performance of student such as students' study habits, study period, result and others. Otherwise, this dataset has a mix of nominal, ordinal, interval and ratio data that offering a comprehensive foundation for applying various statistical analysis techniques. Thus, we want to evaluate whether these variables affect the students' academic performance.

The 2-sample hypothesis test, correlation test, regression test and chi-square test for independence are used to test the sample data. We are using RStudio for data analysis step to generate a graphical presentation of data and do some of the basic calculation.

3.0 Data Analysis

3.1 Hypothesis 2 sample test

Based on the data we obtained, we wish to determine if there is any difference in mean of study hours between student who study regularly and those who study seldom. So that the hypothesis tests on mean for 2 independent samples are carried out and the population variances are unknown. The sample size of student who study regularly is $n_1 = 17$ while for student study seldom $n_2 = 52$. We will use F test to determine the population variance for both group are equal or unequal.

Statistical Inferences for Two Sample Test on Standard Deviation:

Hypothesis statement:

Null hypothesis, $H_0 : \sigma_1^2 = \sigma_2^2$

Alternative hypothesis, $H_1 : \sigma_1^2 \neq \sigma_2^2$

Where σ_1^2 represent the population variance study hours for student who study regularly while σ_2^2 represent population variance study hours for student who study seldomly.

$$F = \frac{S_1^2}{S_2^2}$$

```
> Fcritical <- qf(1-0.05,51,16)
> print(Fstatistic)
[1] 2.333769
> print(Fcritical)
[1] 2.121876
>
> #to identify it is equal or unequal variance , using F test
> if (Fstatistic>Fcritical){
+   print("This two sample is Assumed Unequal")
+ } else if(Fstatistic<Fcritical){
+   print("This two sample is Assumed Unequal")
+ } else {
+   print("This two sample is Assumed Equal")
+ }
[1] "This two sample is Assumed Unequal"
```

From the calculation in R , we assumed that the population variance for the both group are unequal. Then the test statistic formula and the degree of freedom are shows at below:

$$\text{Test statistic formula, } T_0^* = \frac{\bar{X}_1 - \bar{X}_2 - \Delta_0}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}; \text{ degree of freedom, } v = \frac{(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2})}{\frac{(\frac{S_1^2}{n_1})}{n_1 - 1} + \frac{(\frac{S_2^2}{n_2})}{n_2 - 1}}$$

where,

$\bar{X}_1$ = sample mean of first sample

$\bar{X}_2$ = sample mean of second sample

S_1^2 = sample variance of first sample

S_2^2 = sample variance of second sample

n_1 = sample size of first sample

n_2 = sample size of second sample

Inferences About Two Means: Independent Sample (Unequal variance)

Hypothesis statement:

Null hypothesis, $H_0 : \mu_1 = \mu_2$

Alternative hypothesis, $H_1 : \mu_1 \neq \mu_2$

Where μ_1 represent the population mean study hours for student who study regularly while μ_2 represent population mean study hours for student who study seldomly.

Test Statistic:

The sample mean of study hours for student who study regularly, $\bar{X}_1 = 3.588$, while sample mean of study hours for student who study seldomly, $\bar{X}_2 = 2.731$. The sample standard deviation for the both sample are calculated in R. The results are $S_1^2 = 2.181$ and $S_2^2 = 3.332$. A significant level of 0.05 ($\alpha = 0.05$) is used to test the claim that there is a difference in mean of study hours between student study regularly and seldomly. From the RStudio we obtain the test statistic value, $T_0^* = -0.994$ and the degree of freedom is $df = 42.049$.

```

> print("T statistical value : ")
[1] "T statistical value : "
> print(t0)
[1] -0.9937139
>
> print("T critical value : ")
[1] "T critical value : "
> print(t.alpha)
[1] -2.018082
>
> if (t0>t.alpha){
+   if(t0<-t.alpha){
+     print("Fail to reject H null hypothesis")
+   }else{
+     print("H null hypothesis rejected")
+   }
+ }else if(t0<-t.alpha){
+   if(t0<-t.alpha){
+     print("Fail to reject H null hypothesis")
+   }else{
+     print("H null hypothesis rejected")
+   }
+ } else {
+   print("H null hypothesis rejected")
+ }
[1] "Fail to reject H null hypothesis"

```

Conclusion:

Since $-T_{0.025,42.049} > T_0^* < T_{0.025,42.049}$, thus fail to reject null hypothesis, H_0 . There is no sufficient evidence to support the claims that there is a difference in mean of study hours between student who study regularly and seldomly. We cannot conclude that a student's study hours depend on whether they study regularly or seldomly. The study status does not appear to have a clear impact on the amount of time students spend studying.

3.2 Correlation Test

In the correlation test, we examine the relationship between the study period in a day and CGPA of students in a sample size of 69. According to our dataset, both of our data are interval type. Therefore, we select Pearson's technique to calculate the sample correlation coefficient, r . Below are the formula for calculate the sample coefficient, r :

$$\text{Sample correlation coefficient, } r = \frac{\sum xy - \frac{(\sum x \sum y)}{n}}{\sqrt{\left[\left(\sum x^2 \right) - \frac{(\sum x)^2}{n} \right] \left[\left(\sum y^2 \right) - \frac{(\sum y)^2}{n} \right]}}$$

$$\text{Test statistic formula, } t = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}}; \text{ degree of freedom, } df = n-2$$

where,

r = Sample correlation coefficient

n = Sample size

x = Value of the independent variable

y = Value of the dependent variable

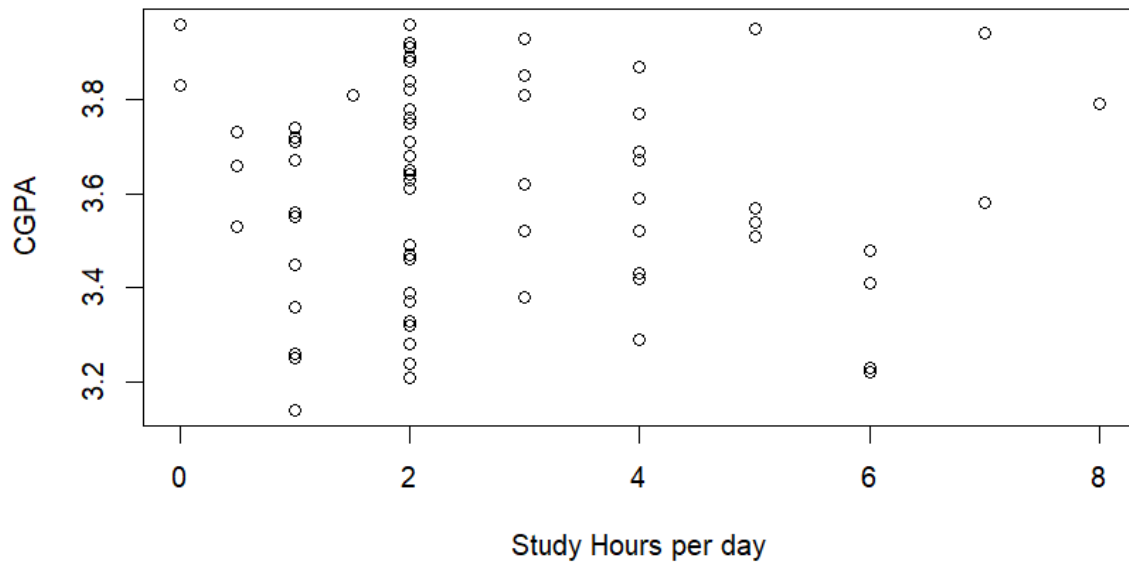


Figure 3.2.1: Scatterplot relationship between CGPA and Study hours per day

From the scatter plot in Figure 3.2.1, we observe that the data set shows a negative linear correlation between CGPA and study hours per day.

Hypothesis statement:

Null hypothesis, $H_0: \rho = 0$ (no linear correlation)

Alternative hypothesis, $H_1: \rho \neq 0$ (linear correlation exists)

where ρ represent the population correlation coefficient.

Test statistic:

A significant level of 0.05 ($\alpha = 0.05$) is used to test the claim that there exists linear correlation between students' CGPA and study hours per day. From the RStudio we can calculate the correlation coefficient by using `cor()` and get the value $r = -0.0411$, test statistic value, $t = -0.337$ and the degree of freedom, $df = 67$. The probability value that we obtain is $P\text{-value} = 0.7371$.

Conclusion:

Since $P\text{-value} > \alpha$ ($0.7371 > 0.05$), thus fail to reject null hypothesis, H_0 . There is not enough evidence to reject the claims that there exists linear correlation between students' CGPA and their study hours per day. As everyone having different study mode, it doesn't mean that study using longer time will get a better score. There are also people that able to score good marks without studying with longer time. Therefore, student must choose the suitable study mode to perform a more efficient study.

3.3 Regression Test

In the regression test, we measure the relationship between the number of courses taken by the students and students' comfortability of study mode with the sample size, $n = 69$. The independent variable, x is Number of Courses Taken by Students and the dependent variable, y is Students' Comfortability of Study Mode. The sample regression line provides an estimate of the population regression line.

Estimated Regression Model, $\hat{y}_i = b_0 + b_1x$

where,

$\hat{y}_i$ = estimated dependent variable value

b_0 = estimated of the regression intercept

b_1 = estimated of the regression slope

x = independent variable

Test statistic formula, $t = \frac{b_1 - \beta_1}{s_{b_1}}$; degree of freedom, $df = n - 2$

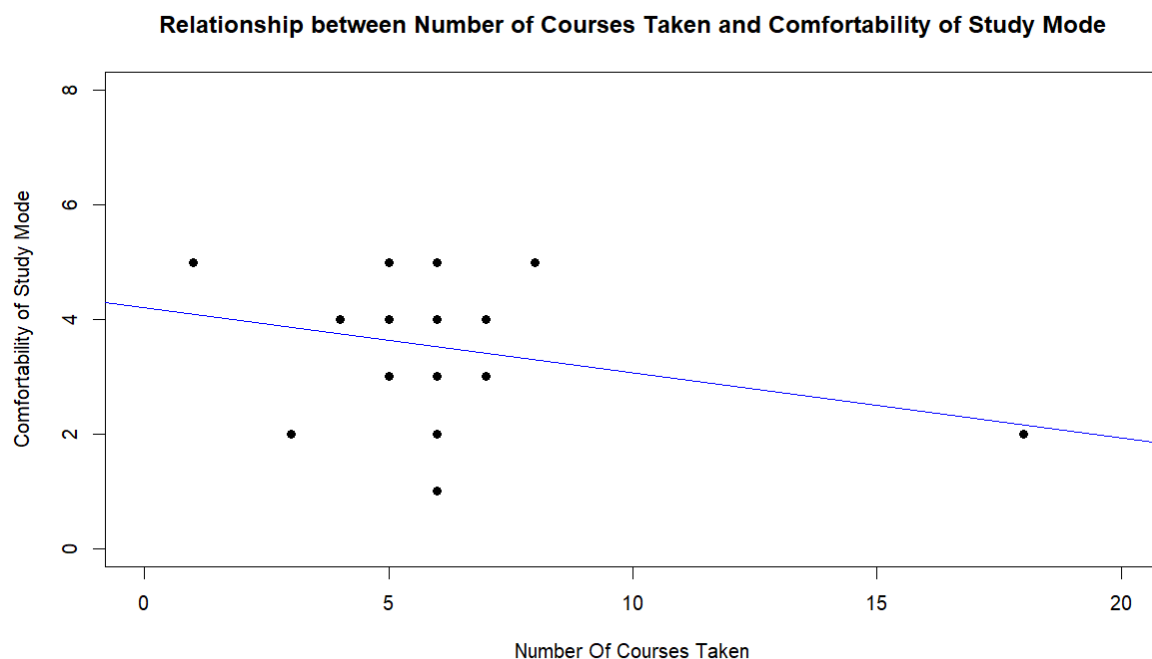


Figure 3.3.1: Scatterplot relationship between Number of Courses Taken and Comfortability of Study Mode

From the scatter plot in Figure 3.3.1, we can see that there is a negative linear relationship between Number of Courses Taken and Comfortability of Study Mode. The best fit line shows a downward inclining.

From the summary, we can the formula for estimated regression model which is

$$Y = 4.2049 - 0.1139x$$

From the formula, the value of $b_0 = 4.2049$ is the estimated value of y when the value of x is equal to 0. While the value of $b_1 = -0.1139$ measures the estimated change in the average value of comfortability of study mode, y as a result of one-unit change in number of courses taken, x .

The coefficient of determination is the portion of the total variation in the dependent variable that is explained by variation in the independent variable.

$$R^2 = \frac{SSR}{SST} = \frac{\text{sum of squares explained by regression}}{\text{total sum of squares}}$$

From the RStudio we obtain the value of coefficient of determination, $R^2 = 0.036$.

Since $0 < R^2 < 1$, the result shows a weaker linear relationship between the number of courses taken, x and comfortability of study mode, y . Some but not all of the variation in comfortability of study mode, y is explained by the variation in number of courses taken, x .

Hypothesis statement:

Null hypothesis, $H_0 : \beta_1 = 0$ (no linear relationship)

Alternative hypothesis, $H_1 : \beta_1 \neq 0$ (linear relationship does exist)

Where β_1 represent the population slope coefficient (hypothesized slope).

Test statistics:

A significant level of 0.05 ($\alpha = 0.05$) is used to test the claim that there exists a linear relationship between the number of courses taken and comfortability of study mode for each student. From the RStudio we obtain the sample regression slope coefficient, $b_1 = -0.1139$, estimator of the standard error of the slope, $s_{b_1} = 0.0722$ test statistic value, $t = 1.576$ and degree of freedom is $df = 67$. The probability value that we obtain is $P - \text{value} = 0.1196$.

Conclusion:

Since $P - \text{value} > \alpha$ or ($0.1196 > 0.05$), thus fail to reject null hypothesis, H_0 . There is not sufficient evidence to support the claim that there exists a linear relationship between the Number of Courses Taken and Comfortability of Study Mode for each student. We can see the residual and residual standard error which is the difference between y and predicted y is larger, R^2 is low, so there is no strong linear relationship. But we can make the conclusion that there is least significant relationship between number of courses taken and comfortability of study mode because we just prove that there is a linear relationship from the scatter plot.

3.4 Chi-square Test of Independence

Chi-square test of independence is used to test for the relation between two nominal variables. To proceed with the Chi-square test, we summarize our data using a two way contingency table. In this test, we test whether there is a relationship between students' study habit and their satisfaction of their learning mode . The formula of test statistic is:

$$\chi^2 = \sum_{all\ cells} \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

Degree of freedom, $df = (r - 1)(c - 1)$

Where,

o_{ij} = Observed count at row i column j

e_{ij} = Expected count at row i column j

r = total number of rows

c = total number of columns

Hypothesis statement:

Null hypothesis, H_0 : students' satisfaction of current learning mode are independent on their study habit.

Alternative hypothesis, H_1 : students' satisfaction of current learning mode are dependent on their study habit.

Test Statistic:

A significant level of 0.05 ($\alpha = 0.05$) is used to test the claim that students' satisfaction of current learning mode are dependent on their study habit. The below show the two way contingency table created by using RStudio.

	Not satisfied	Neutral	Satisfied
regularly	3	5	9
Seldom	8	17	27

Observed frequency table

	Not satisfied	Neutral	Satisfied
regularly	2.710145	5.42029	8.869565
Seldom	8.289855	16.57971	27.130435

Expected frequency table

Pearson's Chi-squared test

```
data:  tbl2
X-squared = 0.086924, df = 2, p-value = 0.9575
```

From the RStudio we obtain the test statistic value, $X^2 = 0.087$ and the critical value X^2 calculated in R is 5.991 .for and the degree of freedom is $df = 3$. The probability value that we obtain is P- value = 0.9575.

Conclusion:

With decision rule: If $X^2 > 5.991$, reject H_0 , otherwise , do not reject H_0 . Since our $X^2 < 5.991$, therefore fail to reject H_0 . There is no sufficient evidence to support the claims that there is a relationship between students' study habit and their satisfaction of their learning mode . So, we cannot be sure that the satisfaction of student learning mode is highly dependent on their own study style, each person's level of satisfaction depends on a variety of factors. It may be that too much study will makes them bored, or that too little study does not cause them anxiety.

4.0 Conclusion

The results of the two-sample hypothesis test show that there is insufficient evidence to support the claim that there is a difference in average study time between students who study frequently and those who study rarely. Therefore, students' study time does not seem to be significantly affected by whether they study often or rarely.

Similarly, the results of the correlation tests showed that there was insufficient evidence for a linear correlation between students' CGPA and the number of hours studied per day. This means that longer study hours do not necessarily lead to better academic performance. Students may achieve high scores through different study habits, emphasizing the need for individuals to find the most efficient way to study.

In regression tests, the results showed no significant linear relationship between the number of courses taken and the comfort level of the learning mode for each student. Although the scatter plot shows an underlying relationship, low R^2 values and high residuals indicate that any such relationship is weak and not strongly linear.

Finally, the Chi-square test results show that there is insufficient evidence to support a relationship between students' study habits and their satisfaction with learning patterns. Satisfaction with learning patterns appears to be influenced by a variety of factors, not just learning habits, suggesting that personal preferences and other external factors play an important role.

In summary, the statistical tests collectively show that study habits, study time, and number of courses taken do not have a direct or strong linear relationship with academic achievement, learning comfort, or satisfaction with learning patterns. Each student's learning experience is influenced by multiple factors, highlighting the importance of personalized and adaptive learning strategies.

5.0 Appendix

Sample of Raw data set

Study habit	Study Hour	CGPA	Number of Courses Tak	Comfortability of study mode	Satisfaction level for learning mode
Seldom	2	3.92	6	2	Not Satisfied
Seldom	4	3.67	6	5	Satisfied
Seldom	1	3.45	5	3	Neutral
Seldom	2	3.28	6	2	Not Satisfied
regularly	2	3.78	6	2	Not Satisfied
Seldom	1	3.14	6	3	Neutral
Seldom	3	3.52	6	3	Neutral
Seldom	2	3.89	6	4	Satisfied
Seldom	1	3.72	6	2	Not Satisfied
Seldom	2	3.61	6	2	Not Satisfied
Seldom	1	3.55	6	5	Satisfied
regularly	3	3.81	6	4	Satisfied
Seldom	2	3.37	6	4	Satisfied
Seldom	2	3.64	6	5	Satisfied
Seldom	5	3.95	5	4	Satisfied
Seldom	2	3.49	6	3	Neutral
Seldom	1	3.26	6	5	Satisfied
Seldom	0.5	3.73	1	5	Satisfied
Seldom	7	3.58	6	5	Satisfied
Seldom	2	3.68	5	4	Satisfied
Seldom	0	3.83	6	5	Satisfied
Seldom	2	3.21	6	3	Neutral
Seldom	4	3.42	6	3	Neutral
Seldom	2	3.75	6	4	Satisfied
regularly	2	3.32	6	3	Neutral
Seldom	1	3.67	7	3	Neutral
Seldom	2	3.88	6	3	Neutral
regularly	1	3.56	6	3	Neutral
Seldom	4	3.29	6	4	Satisfied
Seldom	2	3.96	6	3	Neutral
regularly	1	3.71	6	4	Satisfied
Seldom	2	3.47	6	3	Neutral
Seldom	2	3.63	6	3	Neutral
regularly	2	3.84	6	4	Satisfied
Seldom	2	3.39	6	4	Satisfied
regularly	4	3.77	6	4	Satisfied
Seldom	4	3.52	7	4	Satisfied

Data Resource Link

[DataSet.xlsx](#)

Video Link

<https://youtu.be/daq42ywWTRc>

